SA06 B PROVIDE CLEAN AIRFLOW RATES FOR CONTROL OF INFECTIOUS AEROSOLS | O (MAX: 2 PT)

Intent: Minimize the exposure risk to respirable infectious agents.

Summary: This WELL feature requires projects to provide sufficient clean airflow rates to reduce the risk of airborne disease transmission.

Issue: When humans exhale, they can release pathogens, such as COVID-19 or Influenza particles, which can remain suspended in the air for several hours or longer. These pathogens can be recirculated through ducts and HVAC systems of buildings, increasing the risk of transmission of disease indoors.

Solutions: The simplest way to avoid recirculating contaminated air is to not recirculate it by supplying spaces with 100% outdoor air. Unfortunately, in certain climates this can result in high energy usage, although this can be mitigated through the use of heat recovery systems.[9] In buildings where recirculated air is utilized, it can be treated to remove contaminants. Carbon is capable of filtering VOCs and ozone from the air that passes through[10,11]. HEPA or near-HEPA filters can help remove virus particles, since the virus often travels as an attachment to a larger particle.[12,13] UVGI systems can also be effective, both when irradiating the upper portion of the room or when installed in the air ducts, so long as they are powerful and/or the air speed is slow enough to provide a sufficient UV dose.[13,14] Other technologies exist, but should be evaluated carefully. For example, more novel devices that release oxidants into rooms have less robust evidence for safety and efficacy outside of laboratory conditions.[15,16] Finally, in-room air purifiers can be beneficial because the clean air is often provided within the breathing zone. For optimal performance, air filtration systems need to be maintained according to the manufacturer's instructions.

Part 1 Part 1 (Max: 2 Pt)

For All Spaces:

The mechanical system provides all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:

a. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).[17]
b. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix SA1.[18]

OR

The following requirements are met:

a. The mechanical system is capable of providing all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:
    1. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).[17]
    2. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix SA1.[18]
b. The project has an operational plan for Infection Risk Management Mode (IRMM) that includes the following:[17]
    1. The conditions under which the IRMM will be activated.
    2. The engineering controls (e.g., changes in ventilation or air treatment) and non-engineering controls (e.g., changes in occupancy) that will be utilized to achieve the target clean airflow rates.
    3. The operational and maintenance procedures in place to implement target clean airflow rates within 48-hours of the decision to activate IRMM.
    4. An inventory of HVAC system consumables (e.g., filters, adsorption media, UV bulbs) that is updated when significant equipment modifications have been implemented or after replacement consumables have been utilized.

5. A requirement that there is at least one set of replacement consumables available to install during IRMM.

APPENDIX SA1:

eACH = Q / [room volume][18]

Where Q = A + B + C + D (total clean volumetric air flow rate)

To determine the total clean volumetric air flow rate (Q) for a room, sum the air flow rates for individual systems (as applicable to the HVAC system) per the calculations below.

A: Ventilation Systems[18]

The greater of:

The full mechanical HVAC outdoor air supply rate

The full mechanical HVAC system's exhaust rate, provided the transfer air is from non-regularly occupied spaces or outdoors

B: CADR-rated Media Filters (Standalone Systems)[17,18]

One of the following:

The full m-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-5) or, if unavailable,

The full smoke-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-1)

C: In-duct or non CADR-rated Media Filters[18]

Multiply the air supply rate (e.g., mechanical HVAC fan, blower fan) by the weighted clean air below:

MERV-A (ASHRAE 52.2)	ePM2.5 (ISO 16890)	Weighted Clean Air Factor (MERV E2 [1-3 um] minimum arrestance efficiency)
< 13	≤ 79	0
13	80	85
14-15	85	90
16	90	95
17-19 (HEPA	99	99

D: In-duct UVGI[17]

Multiply the air supply rate by the infectious aerosol reduction efficiency determined in accordance with ANSI/ASHRAE 185.1 with MS2 challenge organism.